



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 40 summarizes the current fairness state for upload 40. Accuracy changed to +0.964 and fairness score changed to +0.897. Strong fairness improvement with no measurable accuracy loss. The best balanced model is Pipeline_mitigated_reweighting with combined score 0.924.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.964 ▲ +0.964	0.897 ▲ +0.897	0.924 Pipeline_mitigated_rewei ghting	0.366 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	dt_baseline.pkl	comparison

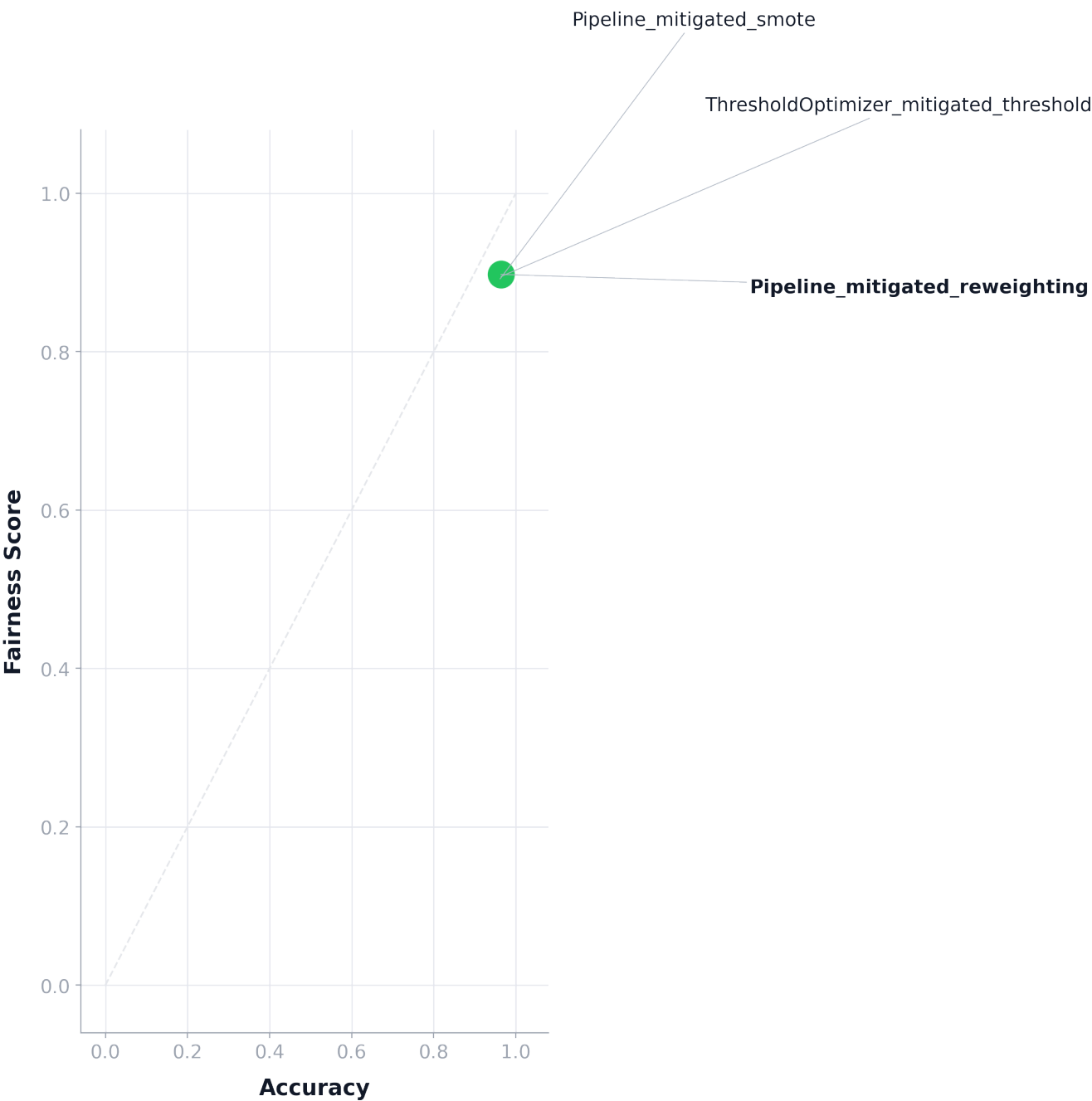
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.193	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.012	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.366	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ Pipeline_mitigated_reweighting	0.964	0.897	0.193	0.012	0.366	0.924
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ThresholdOptimizer_mitigated_threshold	0.964	0.895	0.193	0.018	0.366	0.923
Pipeline_mitigated_smote	0.962	0.892	0.195	0.022	0.370	0.920

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.